



**TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
THAPATHALI CAMPUS**

**A MINOR PROJECT REPORT
ON
AUTOMATED NUTRIENT DOSING AND MONITORING SYSTEM FOR
HYDROPONIC CROP PRODUCTION**

Submitted By:

Aradhya Dhungel	(THA079BEI005)
Dikshya Khatri	(THA079BEI010)
Nischal Dangol	(THA079BEI016)
Pankaj Kumar Mehta	(THA079BEI018)

Submitted To:

Department of Electronics and Computer Engineering
Thapathali Campus
Kathmandu, Nepal

In partial fulfillment for the award of the
Bachelor's Degree in Electronics, Communication and Information Engineering

Under the Supervision of

Er. Anup Shrestha

March 2026



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DECLARATION

We hereby declare that the project entitled, “**Automated Nutrient Dosing and Monitoring System for Hydroponic Crop Production**” in the partial fulfillment of the requirements for the award of the Degree of Bachelor of Engineering **Electronics, Communication and Information Engineering** is a bona fide report of the work carried out by us. These materials contained within the report have not been submitted to any University or Institution for the award of any degree, and we are the only authors of this complete work and no sources other than the listed ones have been used in this work.

Aradhya Dhungel (THA079BEI005) _____

Dikshya Khatri (THA079BEI010) _____

Nischal Dangol (THA079BEI016) _____

Pankaj Kumar Mehta (THA079BEI018) _____

Date: March 2026

CERTIFICATE OF APPROVAL

The undersigned certify that they have read and recommended to the **Department of Electronics and Computer Engineering, IOE, Thapathali Campus**, a minor project work entitled “**Automated Nutrient Dosing and Monitoring System for Hydroponic Crop Production**” submitted by **Aradhya Dhungel, Dikshya Khatri, Nischal Dangol, Pankaj Kumar Mehta** in partial fulfillment for the award of Bachelor of Engineering in Electronics, Communication and Information Engineering. The project was carried out under the special supervision and within the time prescribed by the syllabus.

We found the students to be hardworking, skilled and ready to undertake any related work to their field of study and hence we recommend the award of partial fulfillment of Bachelor’s Degree of Electronics, Communication, and Information Engineering.

Project Supervisor

Er. Anup Shrestha

Department of Electronics and Computer
Engineering

Thapathali Campus, IOE

External Examiner

Dr. Ritu Raj Lamsal

MTech, PhD, FIE

Department of Digital Technology

MBUST, Nepal

Head of Department

Er. Umesh Kanta Ghimire

Department of Electronics and Computer
Engineering

Thapathali Campus, IOE

March 2026

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Aradhya Dhungel	(THA079BEI005)
Dikshya Khatri	(THA079BEI010)
Nischal Dangol	(THA079BEI016)
Pankaj Kumar Mehta	(THA079BEI018)

ABSTRACT

This project demonstrates the design and implementation of an automated hydroponic system. It allows soilless growth of crops for efficient resource consumption and improved yield and productivity as compared to traditional farming. Hydroponics refers to the water-based cultivation method in which plants are grown in a solution with calculated nutrient quantities for individual crops. The designed system consists of sensors such as pH sensor and EC sensor, an ESP32-WROOM32 microcontroller-based control unit, and automated actuators to continuously monitor and regulate key parameters such as pH, EC, temperature and humidity. Automation of the nutrient solution is done through a closed-loop control of pH and EC of the solution where the sensors provide real-time feedback to the controller, which compares measured values with predefined crop-specific standard EC and pH ranges. Based on these parameters, dosing pumps are used to add required quantities of acid, base, or concentrated nutrient solutions to compensate for deviations in the nutrient-rich solution. Calibration routines are performed at regular intervals to ensure sensor accuracy and prolonged system reliability. It reduces pH deviation by 50 – 68% and stabilizes EC within $\pm 30 - 60$ ppm range.

Keywords — *calibration, closed loop algorithms, dosing, ESP32, EC, hydroponic, pH, TDS*

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LIST OF ABBREVIATIONS

AC	Alternating Current
ADC	Analog To Digital Converter
API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
DC	Direct Current
DRF	Django Rest Framework
EC	Electrical Conductivity
ESP	Espressif Systems Programmable
GPIO	General Purpose Input/Output
HTTP	HyperText Transfer Protocol
I ² C	Inter Integrated Circuit
IoT	Internet of Things
JWT	JSON Web Token
pH	Potential of Hydrogen
PID	Proportional Integral Derivative
ppm	Parts Per Million
RH	Relative Humidity
TDS	Total Dissolved Solids

1. INTRODUCTION

1.1. Background

In this modern age, rapid population growth, climate change, degradation in the quality of soil, and declining availability of arable land have placed significant pressure on conventional agricultural practices. Considering this fact, hydroponic farming is used to cultivate crops in vertically stacked layers instead of traditional horizontal farming methods.

In hydroponics, we externally provide nutrients that plants require to grow and develop. Nutrients needed for the plant are mixed within the water and thus directly provide the nutrients to the plant roots. All plants will get an equal share of nutrients, which leads all plants to grow equally. Hydroponics started to flourish in the early twentieth century, when William Frederick Gericke of the University of California popularized the idea of growing crops in nutrient solutions rather than soil for large-scale production. The reason behind hydroponic farming is to maximize production in limited spaces, higher crop yield, and faster growth.

As technology has become vital and more accessible, our project aims to automate hydroponic systems where temperature, humidity, pH, and EC can be monitored and controlled. These parameters are essential for maintaining optimal growing conditions and ensuring healthy plant growth. Among all the parameters, pH and EC are prioritized as plant growth depends directly on the amount of micro-nutrients and macro-nutrients which EC and pH directly control the availability and absorption of the nutrients. Integration of IoT in the system can help reduce human intervention through automatic nutrient dosing using pumps, and monitor nutrient levels in real time using sensors. The data collected by the sensor is then processed by a microcontroller which determines if the concentration of nutrients is within the range for a healthy plant growth or not. Users can track nutrient consumption, parameters, and dosing times through the web based interface.

In this context, the project focuses on the design and implementation of an automated hydroponic system in order to provide accurate nutrient delivery. The designed approach minimizes human error, reduces nutrient wastage, and sustains small-scale indoor agricultural applications.

1.2. Motivation

In a densely populated city facing rapid urbanization, such as Kathmandu, the toll taken on traditional farming has been massive. With less and less arable land, water scarcity, and lower crop yield with time, conventional soil-based farming is facing increasing

challenges and turning into a struggle to keep up with. Hydroponics mitigates such drawbacks since it does not require a big space, does not require soil, is free from the majority of disease-causing parasites, circulates required amounts of water and nutrients, thus promoting water recycling and higher yield. We therefore determined that an automated nutrient-dosing hydroponic system was the most viable approach to modernize farming with minimal drawbacks.

Hydroponics, even done manually, is a highly effective and sustainable approach to cultivation. Introducing automation to it reduces time losses, cuts down human errors, and boosts the accuracy, precision, and overall performance of the model. This ultimately leads to high crop yield, high production, and less water use, all with minimal effort and fewer challenges through smart management of resources.

1.3. Problem Definition

Hydroponics requires constant monitoring and utmost precision, which is difficult to achieve through manual labour, but is generally done manually. Parameters such as pH, EC, and humidity differ from plant to plant, which must be provided with a minimum tolerance. Even small deviations, if unnoticed for a longer time, which is very likely with manual models, can cause nutrient deficiencies, root stress, and ultimately crop loss. Also, in a manual system, logs and feedback in realtime are unavailable. All the mentioned cases affect the overall efficiency of crop production. Although automation in hydroponics has been around for sometime, the existing systems lack stability aware dosing methods and threshold based control strategies which is also suitable for low cost embedded systems.

1.4. Objectives

- To design and implement a fully automated closed loop hydroponic system.
- To ensure precise realtime remote monitoring and control of key parameters.

1.5. Scope and Application

- The system is designed for a single crop which is lettuce to ensure precise control during experimentation.
- Provides accurate nutrient distribution for each crop through a closed loop pH and EC control system.
- Minimizes human interaction through automated pumps, and integrated sensors which also lowers errors.
- Applicable in areas with land scarcity or/and water.

- Applicable for small farms, home systems, educational institutions as it is affordable as well as scalable.
- Effective use of water and nutrients for sustainable farming.

1.6. Report Organization

This report consists of eight main chapters:

1.6.1. Introduction

It includes the background of the project, problem statement, motivation behind the study and work, objectives, and scope and applications.

1.6.2. Literature Review

It reviews existing literature in hydroponic systems, automation techniques, and IoT based monitoring solutions, along with their advantages and limitations.

1.6.3. Requirement Analysis

It provides the functional and nonfunctional requirements of the system, as well as hardware requirements for the project.

1.6.4. Methodology

It explains the overall system design, system architecture, sensor subsystem, control strategies, pump mechanisms, user interface, and working flow of the system.

1.6.5. Implementation Details

It describes the hardware and software implementation of the system, including sensor interfacing, ESP32 firmware, backend, database, and frontend development.

1.6.6. Result and Analysis

It presents the experimental results obtained from the system such as visualization of sensor data, evaluation metrics, and analysis of the obtained results.

1.6.7. Future Enhancements

It discusses the limitations as well as the possible improvements that can be made in the future.

1.6.8. Conclusion

It summarizes the overall work, results, and final outcomes of the project. Also, the report includes Appendices (budget, schedule, circuit diagrams, PCB design) and References used throughout the project.

2. LITERATURE REVIEW

Hydroponics has gone under significant development since it was first introduced. The concept of a soilless farming method where plants can grow with their roots immersed in a nutrition solution, was established by Gericke's work in 1937, which presented the commercial possibility of "aquaculture" [1]. This work led to the foundation for modern Controlled Environment Agriculture(CEA).

After that hydroponics has evolved hugely and has been adopted in greenhouses and indoor planting. As CEA integrated environmental and nutrient control, and automation, the limitations of traditional soil based farming has been reduced. Savvas and Gruda [2] emphasized that soilless farming technologies play an important role in modern greenhouse production due to its higher crop productivity and less dependency on the quality of soil.

The main advantage of the hydroponics system is that it is far more efficient with resources than traditional farming methods. Barbosa et al. [3] conducted an analysis of lettuce production comparing the hydroponic and conventional farming methods. The results showed that hydroponics can reduce water consumption by 70 - 90% as it recirculates the nutrient solution. This significantly minimises losses from evaporation and percolation. As the nutrients are directly exposed to the roots, fertilizer use is efficient and nutrient draining into the environment is less. This addresses the major ecological concerns related to traditional agriculture [2].

Another major advantage of hydroponics is the efficiency of nutrient use is improved. In conventional farming, nutrients can become unavailable due to fixation, leaching, or microbial competition. Compared to that, hydroponics provides the nutrients directly to the roots in the form of soluble ions. This provides better control over the concentration of the nutrients [4]. The direct delivery of nutrients reduces wastage of fertilisers ultimately helps to prevent pollution, which is a big problem in conventional agriculture [2].

The critical parameters that affect the performance of hydroponic systems are pH and EC. The availability of nutrients and its solubility depend on the pH level of the solution. Generally, crops grow better with a slight acidic pH range of 5.5 to 6.5 [4]. EC is directly related to the total dissolved solids in the solution, which indicates the nutrient concentration. Keeping these parameters in check within the optimal limits is important to prevent situations such as nutrient lockout or deficiency, which can have a negative impact on the quality and production of the crops [5].

As mentioned earlier, EC is an indirect measure of TDS in the nutrient solution. It

provides information on the estimate of overall concentration of nutrients. Controlling the EC level within the limits defined for specific crops plays a vital role to avoid nutrient deficiencies as well as osmotic stress. If the EC value becomes lower than the threshold, it can lead to insufficient supply of nutrients. Similarly, higher EC values can damage the roots of the plants as well as reduce water uptake. Both high and low EC values negatively impact plant growth and yield [5].

In comparison to soil based farming, nutrient management in hydroponics is more difficult due to the effects of pH and EC on each other. If there are changes in the concentration of nutrients, it may affect the pH level of a solution. Similarly, the balance of nutritional ions can be altered by changes in pH level. Hence, in order to maintain optimal growth and plant development, precise and real-time observation of both EC and pH parameters is required [5].

Traditionally, to monitor and adjust these parameters manual intervention was required which is time consuming, susceptible to human error, and often inconsistent. Early automation efforts primarily focused on timer based nutrient dosing. However, advancements in low cost, reliable sensors and microcontrollers have enabled closed loop feedback control systems. Mendoza et al. [6] demonstrated an automated hydroponic system using Arduino based pH and EC sensing combined with Proportional Integral Derivative (PID) control and peristaltic pumps, resulting in more stable growing conditions compared to manual control. Similarly, Nam et al. [7] compares conventional farming with hydroponics, aeroponics, and aquaponics, emphasizing the critical role automation plays in increasing accuracy, lowering human error, and ensuring consistent root zone conditions. Continuous monitoring and prompt corrective intervention are made possible by automated devices, which are especially crucial for green products like lettuce that develop quickly.

The integration of Internet of Things (IoT) represents the next stage of hydroponic automation. IoT enabled projects support cloud based data storage, provide realtime remote monitoring, and automated warning on anomalies or/and failures. A review by Kumar et al. [8] concluded that IoT based hydroponic systems enhance traceability of errors, provide decision support capabilities to known failures, and facilitate the integration of predictive control techniques, including machine learning models. IoT makes it possible to continuously monitor environmental factors, performance, and nutrient parameters in hydroponic applications without the need for constant presence of any human resource. Long term trend analysis of pH, EC, temperature, and humidity is made possible by data logging, which supports data driven optimization and early anomaly detection. IoT solutions also make it possible to provide alerts when parameters alienate from permitted ranges, which lowers the risk of crop damage and

allows for quick corrective action to compensate for the drift [8].

Despite these advantages, several challenges remain, particularly regarding sensor reliability and sensor drift during prolonged measurement periods. Additionally, the need for addressing flexible and cost effective solutions tailored to small scale, urban, and indoor farming applications is continually rising, as many existing methods are designed for specific crops or large scale commercial use.

Table 2-1. Comparison of Existing Systems with Designed System

S.N.	Reference	System Type	Control Method	Key Features	Limitations	Comparison with designed System
1	Gericke (1937) [1]	Basic Hydroponics	Manual	Introduced concept of soilless farming	No automation, no monitoring system	Full automation and real-time control
2	Savvas & Gruda (2018) [2]	Soilless Cultivation	Manual / Semi - automated	Improved yield and efficiency	Limited automation and control precision	Precise parameter control using sensors
3	Barbosa et al. (2015) [3]	Hydroponic vs Conventional	Not control - focused	High water efficiency (70–90% saving)	No control system implemented	Control and monitoring for efficiency
4	Trejo-Téllez & Gómez-Merino (2012) [4]	Nutrient Solution Study	Manual	Defined optimal pH and nutrient ranges	No real-time monitoring or automation	Maintenance of optimal ranges
5	Hansen (1978) [5]	Nutrient Management	Manual	Explained EC and pH importance	No automation, prone to human error	Elimination of manual adjustment errors

S.N.	Reference	System Type	Control Method	Key Features	Limitations	Comparison with designed System
6	Mendoza et al. (2020) [6]	Automated Hydroponic System	PID Control	Closed-loop control using sensors and pumps	Complex tuning, risk of overshoot, higher computation	Simpler threshold-based control with stabilization delay
7	Nam et al. (1996) [7]	Hydroponic Environment Control	Partial automation	Improved crop growth using controlled environment	Limited parameter control and no IoT integration	IoT monitoring
8	Kumar et al. (2021) [8]	IoT Hydroponic Systems	Varies (general review)	Cloud monitoring, remote access	Lacks optimized control strategy for low-cost systems	IoT+ efficient low-cost control method
9	Designed System	Fully Automated Hydroponic with IoT	Threshold based Closed loop	Real time monitoring, automated dosing, IoT dashboard, stabilization delay	Limited to single crop, frequent calibration	–

3. REQUIREMENT ANALYSIS

3.1. Functional Requirements

There are four critical parameters. They are pH, EC, temperature and relative humidity. The system measures all four parameters in an interval of 10 minutes. As the system is a closed loop system, it must run dosing pumps when pH or EC values deviate from their thresholds. EC correction must be prior to pH correction as EC may have effects on the pH level. After that, a stabilization delay must be introduced after dosing events to prevent nutrient overshoot. Similarly, the system must activate an exhaust fan in case of high relative humidity. The sensor must communicate wirelessly through Wi-Fi and stored into a cloud database. The stored data is then presented on a web dashboard with values and graphs. The users must be able to control the parameter setpoints and pump/fan activation time remotely.

3.2. Non-Functional Requirements

The system must work automatically with minimal human intervention. It must maintain pH within ± 1 unit and EC within ± 200 ppm of their respective setpoints. Corrective response must bring parameters back within limits in under 10 minutes of the detection of deviation. Sensor accuracy requirements are ± 0.1 pH for the pH sensor and $\pm 2.5\%$ for the TDS sensor. The overall system must be cost effective and built mostly using open source software.

3.3. Hardware Requirements

The control unit requires an ESP32-WROOM32 microcontroller which has sufficient GPIO availability for analog sensing without Wi-Fi interference, specifically using pins GPIO32 - 36 and GPIO39. Power management must include a 12V DC adapter and an LM2596 buck converter.

4. METHODOLOGY

4.1. System Overview

The automated hydroponic system is designed to maintain nutrient level and environmental conditions through continuous monitoring and closed loop control. It consists of three sensors which are pH, TDS, and temperature and humidity sensors, an ESP32 based control unit, and cloud based monitoring to regulate pH, EC, and humidity for hydroponic crop cultivation. The entire system is powered using AC supply.

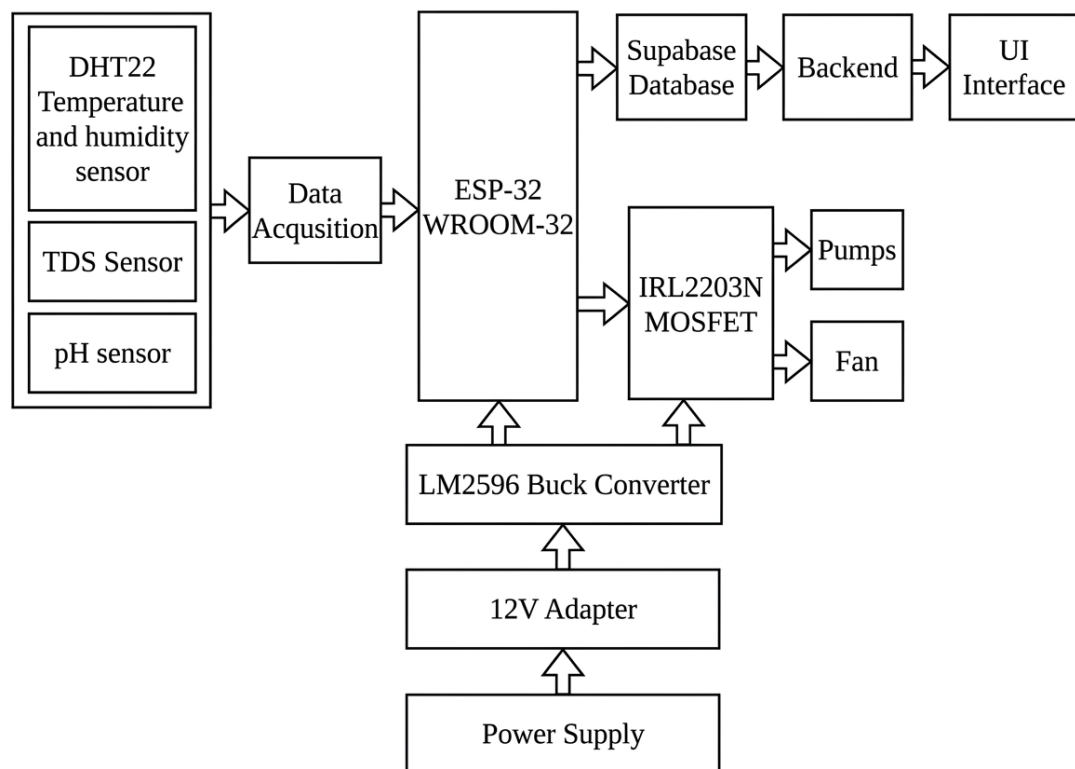


Figure 4-1. Block Diagram of the automated system

The mains AC is first rectified and regulated to 12V by the adapter. The LM2596 Buck step down converter then provides 5V for ESP32 and other peripherals such as pumps. For the sensing stage, three sensors sample the environment in the interval of 10 minutes. The pH sensor converts pH and TDS sensor converts solution conductivity to an analog signal respectively. DHT22 temperature and humidity sensor provides digital output. ESP32 reads the sensor values, compares the current readings with defined setpoints, and decides whether to activate the regulating mechanism. The ESP32's 3.3V GPIO signals drive the gate of an IRL2203N logic level MOSFET, which switches higher current loads (pumps or fan). Three 5V DC dosing pumps handle acid, and two nutrient solutions. A submersible pump recirculates water from the reservoir. Sensor

readings are transmitted through HTTP POST requests to the Supabase cloud database with timestamps which enables remote access and monitoring. The React dashboard displays live sensor readings and historical trend graphs and the user can update pH and EC thresholds remotely.

4.2. Sensor Subsystem

The sensing subsystem is responsible for realtime data acquisition of nutrient parameters, using sensor modules that simplify signal conditioning and communication with the microcontroller.

4.2.1. pH Sensor Module

The pH sensing module, a standard pen type pH meter, continuously monitors the acidity or alkalinity of the nutrient solution over a range of 0 - 14 pHs. The module integrates a glass electrode and a reference electrode with an onboard signal conditioning circuit, providing a voltage output proportional to the pH level. It operates with an input voltage of 3.3V, consuming 5 - 10 mA of current, and outputs a voltage in the range of 2.3 - 0V corresponding to 0 - 14 pH. The module has an accuracy of ± 0.1 pH and a resolution of 0.01 pH, making it suitable for precise nutrient management in hydroponic systems.

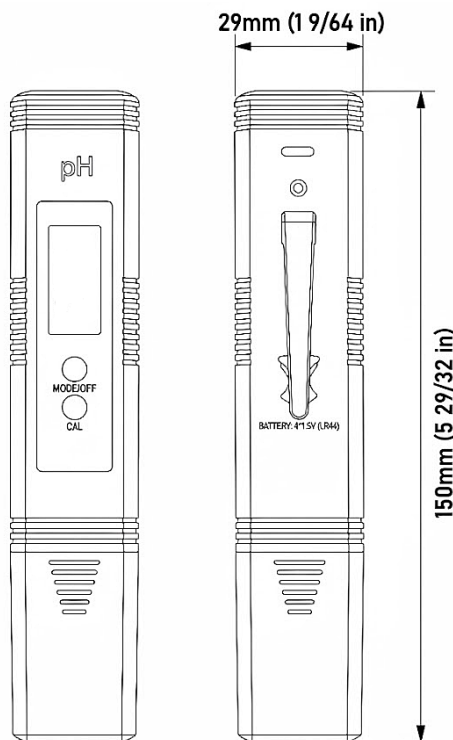


Figure 4-2. Pen type pH Meter

4.2.2. TDS Sensor Module:

The TDS sensing module has been used to measure nutrient concentration by measuring the electrical conductivity (EC) of the solution. It includes temperature compensation circuitry based on ambient temperature within the module itself. The module operates at 3.3V, drawing 3 - 6 mA of current, and provides an analog voltage output of 0 - 2.3V corresponding to 0 - 1000 ppm. The sensor module offers an accuracy of $\pm 2.5\%$ and a resolution of 1 ppm, allowing precise monitoring of nutrient levels in the hydroponic solution. [C-1].

4.2.3. DHT22 Humidity and Temperature Sensor Module

The DHT22 module provides digital measurements of both relative humidity and temperature. It consists of a built in sensor, delivering calibrated readings directly to the microcontroller over a single-wire digital interface. The module operates at 3.3-6V with a working current of 1.5 - 2.5 mA. It measures temperature with an accuracy of ± 0.5 degrees Celcius and a resolution of 0.1 degrees Celcius, and humidity with an accuracy of 2 - 5% RH and a resolution of 0.1% RH. The DHT22 module enables precise monitoring and regulation of environmental conditions.

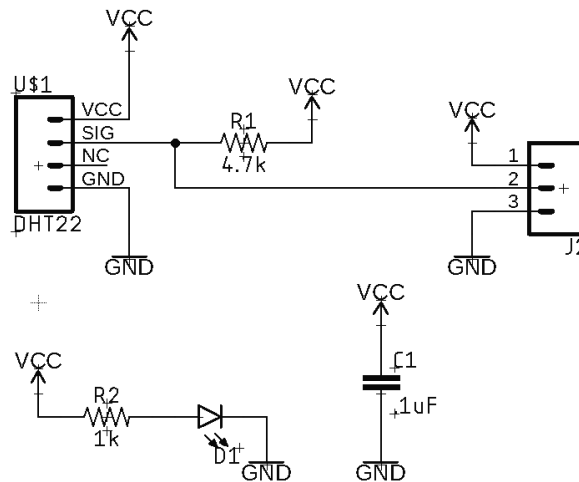


Figure 4-3. Schematic Diagram of DHT22 Sensor Module

4.3. Control and Processing Unit

An ESP32 (ESP32-WROOM32) microcontroller is the central controller [D-1]. Its operating voltage is 3.3V and the maximum rated current for the corresponding development board for both input and output through its GPIO pins is 40mA. Its responsibilities include:

- Processing sensor data via ADC and I²C interfaces

- Executing closed loop control algorithms
- Handling Wi-Fi communication for cloud logging

Crop specific setpoints are stored in its memory and can be updated remotely with the help of the ESP's 2.4Ghz WiFi compatibility.

4.4. Actuation and Nutrient Control

4.4.1. TDS / EC Regulation

- When TDS is below the setpoint macro and micro nutrient solutions are dosed using the pumps in a 3:1 ratio of the solutions.
- EC correction is performed prior to pH compensation to avoid interaction.

4.4.2. pH Regulation

- If measured pH exceeds the upper threshold an acid dosing pump is activated.
- A 5-minute stabilization delay follows each dosing cycle for both pH and EC before remeasurement.

4.4.3. Humidity Control

- If relative humidity exceeds the setpoint exhaust fan gets activated.
- Environmental control is paused during nutrient dosing to ensure measurement stability.

4.5. User Interface and Data Logging

- The readings for pH, TDS, temperature, and humidity sensors are transmitted via Wi-Fi to the backend. Communication is established over Wi-Fi, enabling reliable data transfer between the ESP32 and the server. Authentication is required on the web frontend for enhanced security while allowing realtime monitoring and data logging over the internet.
- The website bridges the gap between human and machine as it enables users to monitor, analyze, and control system parameters efficiently. The website has simple UI/UX design which ensures the critical information can be understood clearly. Sensor readings and logs are stored in a Supabase database, which provides a cloud-based, scalable backend for secure data storage and historical data analysis.
- A dashboard based layout is used to display real time sensor parameters such as pH, EC, temperature, and humidity through visual cards. It also provides control over parameter setpoints i.e sensor threshold values, pump switch ON/OFF time, and exhaust fan switch ON/OFF time. Overall, the UI/UX design enhances user

interaction with the physical counterpart. The calibration inputs are taken from the user and the user is provided with a datasheet including setpoints for each specific crop.

4.6. Other Hardware Components

4.6.1. IRL2203N MOSFET

The IRL2203N MOSFET acts as an electronically controlled switch, allowing low voltage signals from the microcontroller to safely control high-voltage devices. Here, it has been employed to manage the operation of dosing pumps and exhaust fan. It provides reliable switching without the mechanical wear associated with relays or actuators. The MOSFET can handle high currents while being driven by the 3.3V logic output from the ESP32 microcontroller due to its low gate source threshold voltage $V_{GS(th)}$ at 1.0V, making it suitable for our application purposes.

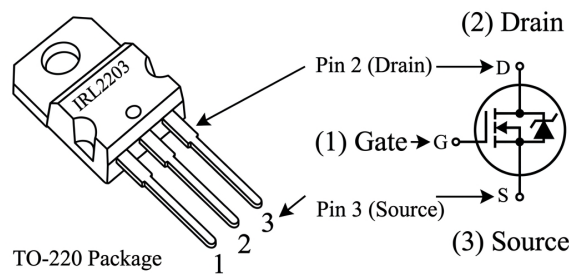


Figure 4-4. IRL2203N MOSFET

4.6.2. Submersible Pump

The submersible pump is used to circulate water from the reservoir to the plant-growing apparatus and back to the reservoir. It operates fully immersed in water, ensuring continuous water flow and proper nutrient distribution. This pump operates on 220 - 240V AC with a current rating of 100 - 120 mA and a power consumption of 25W, providing sufficient flow for our requirements.

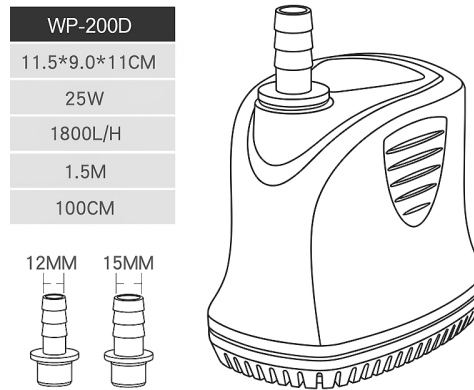


Figure 4-5. AC Submersible Pump

4.6.3. 5V DC Pump

For smaller, controlled transfer of liquid like nutrient and acid dosing, a 5V DC pump is utilized. It operates with an input voltage of 5V DC, drawing between 130 - 220 mA and consuming 2.5W of power. This pump is turned on for fixed durations. With a measured flow rate, we are able to calculate the amount of liquid dosed during the duration of pumping.

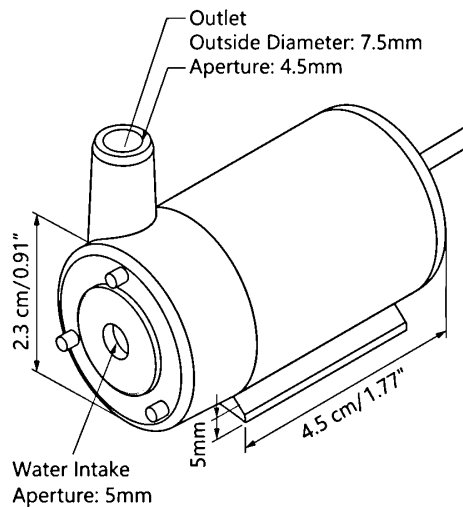


Figure 4-6. 5V DC Pump

4.6.4. AC Power Adapter (12V DC / 1.5A)

An AC power adapter is used to convert mains AC voltage into a stable 12V DC output. This adapter powers pumps and voltage regulation modules. The adapter accepts an input of 220-240V AC and provides a 12V DC output with a maximum current of 1.5A, delivering up to 18W of power

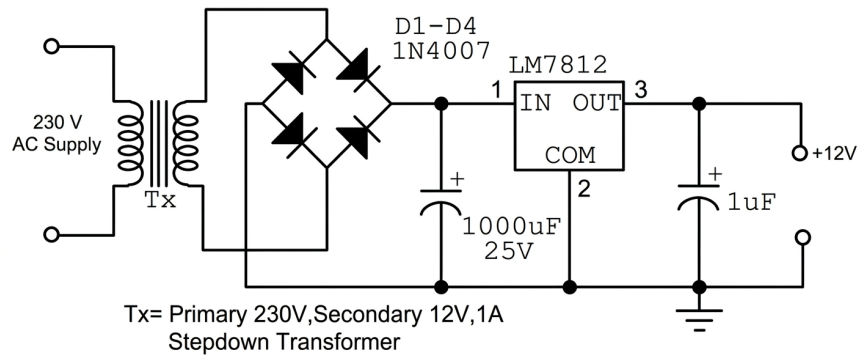


Figure 4-7. AC Power Adapter (12V DC / 1.5A)

4.6.5. LM2596 Buck Converter

To step down higher DC voltages to the lower levels for sensors and control circuits, an LM2596 buck converter is used. It accepts an input voltage range of 4-40V DC and can provide an adjustable output between 1.25 - 35V DC, delivering up to 3A of current and up to 45W of power. This allows stable and efficient voltage regulation for all our electronic components.

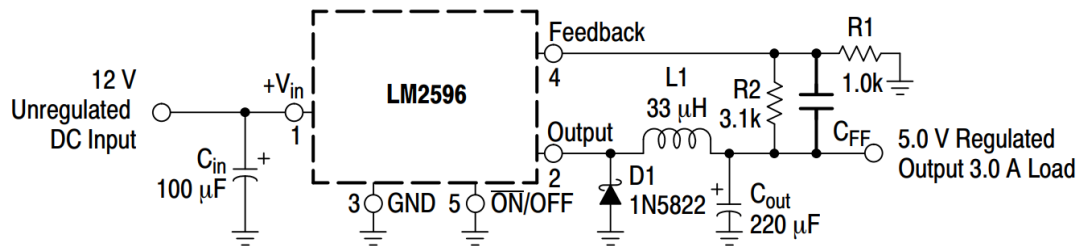


Figure 4-8. LM2596 Buck Converter Module Schematic Diagram

4.6.6. DC Fan

A 12V DC fan is used as an exhaust mechanism to maintain ventilation and heat dissipation for humidity control. The fan draws 0.8 A of current, consuming approximately 9.6W.

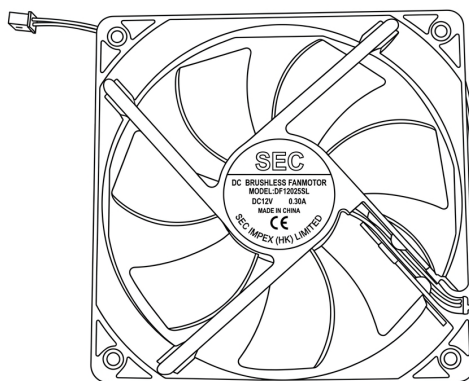


Figure 4-9. 12V DC Fan

4.7. Overall Working Principle

The system operates continuously in a closed loop feedback manner, compensating EC, followed by pH and humidity regulation. Automated calibration has been implemented along with cloud based monitoring to ensure stable long term operation, minimal human intervention, and optimal crop growth conditions.

4.7.1. Experimental Crop

For our model, the experimental crop is lettuce (*Lactuca sativa*). Lettuce is a suitable crop for hydroponic cultivation as it has a short growth cycle, has comparatively low nutrient demand, and is resilient to significant parameter variations in the nutrient solution such as pH and EC. Additionally, the common growth cycle of 1 to 1.5 months creates an opportunity for a proper experimental period within the limited timeframe provided. These mentioned features of lettuce make it an ideal experimental crop for evaluation of the stability and effectiveness of automated nutrient dosing.

As lettuce is already a specialized crop in hydroponics, the growing conditions in manual hydroponics are well-documented in existing studies, allowing contrast and comparison between the experimental results and the standard results.

4.7.2. Experimental Protocol

The experimental procedure of the proposed design is performed using an automated setup which operates over a crop growth cycle of lettuce (*Lactuca sativa*). This design consists of a closed-loop automated control of pH and EC based on the predefined setpoints provided for the crop.

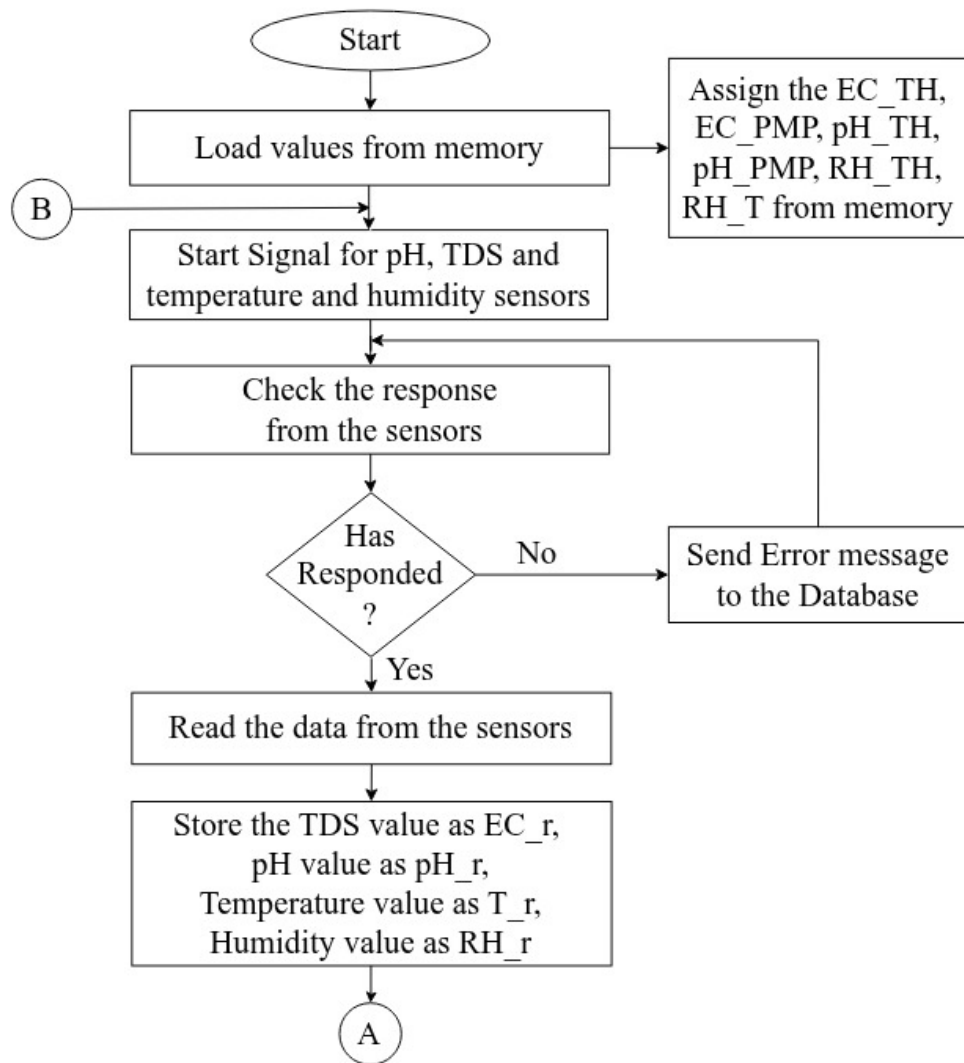
Saplings are first transplanted into the hydroponic system. During the experimental period, the system monitors the nutrient solution parameters in the interval of 10

minutes and autonomously reacts through corrective dosing as per the deviation from the specified range of pH and EC. Furthermore, stabilization delays are applied after each dosing process to allow the nutrients to mix uniformly before the measurements are taken. During each dosing event, the system introduces a priority-based control sequence where EC is balanced before adjusting pH. This is because the concentration of nutrients in the solution influences the acidity of the solution. Conventional IoT based hydroponic systems generally enforce immediate threshold triggered nutrient dosing, where corrective dosing is applied as soon as parameter deviations are detected. In such cases, the delay in mixing is overlooked which causes nutrient overshoot. To solve this problem, a stabilization delay and threshold based control is implemented in the system.

Readings from sensors, dosing events, and nutrient consumption data are monitored and recorded. The collected results are then compared to the reference data published in reports related to manual hydroponics systems cultivating similar crops i.e. lettuce on the basis of parameters such as nutrient consumption, crop yield, pH and EC stability. Benchmark values for pH variation, EC fluctuation, nutrient consumption, and crop growth indicators were sourced from peer reviewed studies and technical reports. Instead of applying PID control, the system utilizes a threshold based proportional dosing mechanism. This control strategy is particularly suited for low-cost embedded platforms and hydroponic systems characterized by slow dynamic response, where robustness and stability are prioritized over real time instantaneous responses

Table 4-1. Comparison of Timer-Based, PID-Based and Designed Control Systems

S.N.	Feature	Timer-Based System	PID-Based System	Designed Control System
1	Control logic	Fixed intervals	Continuous feedback	Threshold-based closed-loop
2	Threshold-based closed-loop	Very low	High	Low
3	Overshoot risk	High	Moderate	Low
4	Sensor dependency	Low	High	Moderate
5	Suitability for low-cost MCU	High	Low	Moderate
6	Adaptability	Poor	High	Moderate



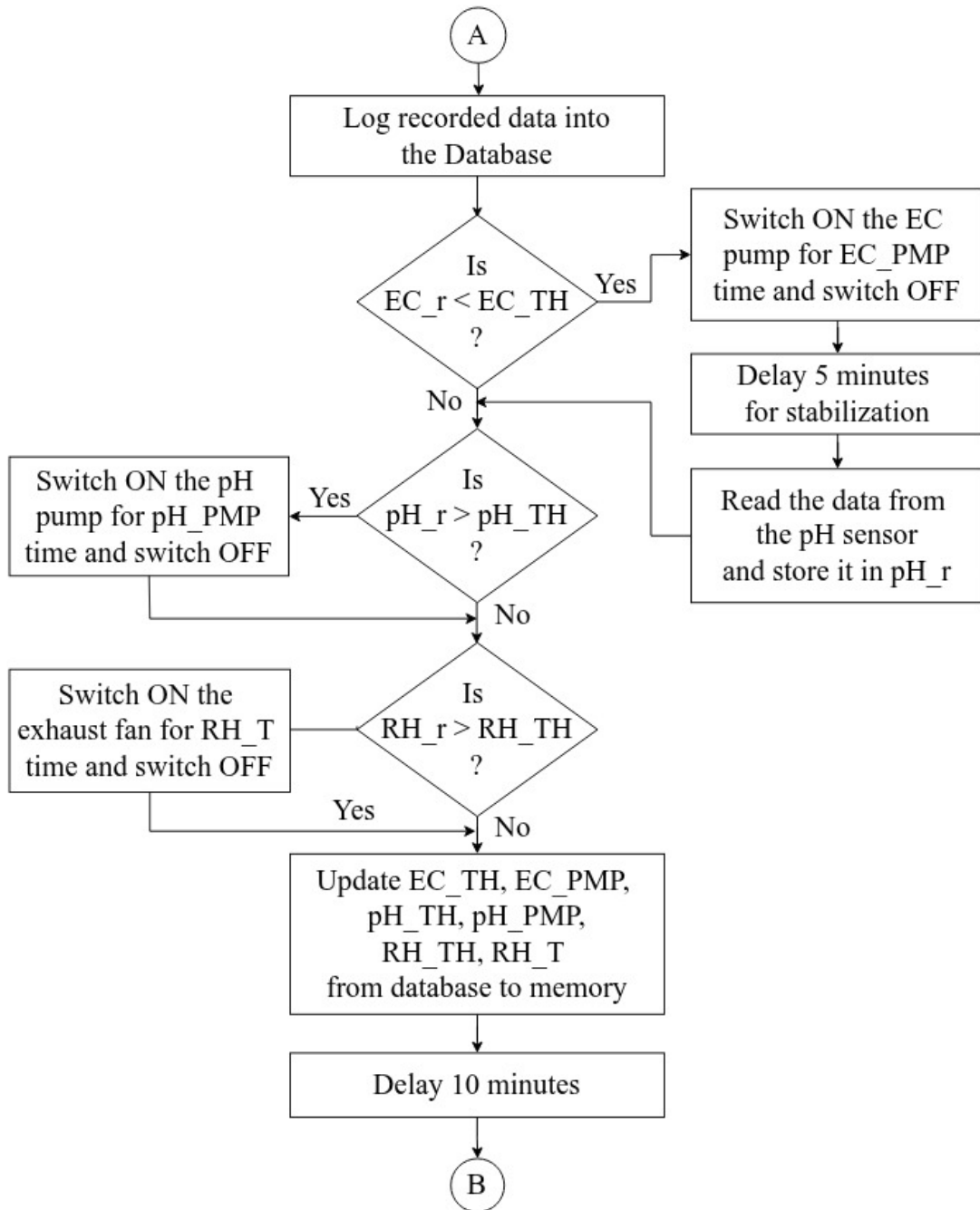


Figure 4-10. Flowchart of the ESP32 Firmware

5. IMPLEMENTATION DETAILS

5.1. Hardware Implementation

5.1.1. pH Sensing (Commercial Pen Type pH Meter)

The pH measurement subsystem consists of a commercial pen type pH meter. Its amplified analog voltage output is connected to the ESP32's GPIO34. GPIO34 is an input only pin which belongs to the ADC1 group, so it remains stable and unaffected during active Wi-Fi operation, which makes it well suited for reliable analog sensing in this system.

Instead of measuring just the raw electrode voltage directly, we performed a three point calibration by immersing the pen type pH meter into standard buffer solutions of pH 4.00, 6.86, and 9.18 provided with the meter. The corresponding amplified voltage outputs from the pH module were recorded at each calibration point. These three voltage - to - pH pairs established a piecewise linear mapping across two segments, one from pH 4.00 to 6.86 and another from pH 6.86 to 9.18, allowing the firmware to interpolate and convert any measured voltage into its corresponding pH value with improved accuracy across the operational range. Linearly interpolating between the data points to calculate constants forming a line of best fit describing the relationship between the electrode voltage and the actual pH of the solutions, the mid_pH i.e. measured electrode voltage at pH 7, was used as such in the following equation to convert voltage readings from the ESP32's ADC to corresponding pH:

$$pH = 7 - \frac{MID_PH_v - pH_raw_v}{CALIBRATION_FACTOR} \quad (5-1)$$

where,

pH = calculated acidity or alkalinity of the solution,

MID_PH_v = reference midpoint value in volts at pH 7,

pH_raw_v = raw sensor reading in volts,

$CALIBRATION_FACTOR$ = constant used for calibration.

The pen-type meter provided a stable, pre-buffered voltage signal to the module, and the ESP32 sampled this through its 12-bit ADC configured with full 3.3V attenuation, yielding a resolution of approximately 0.01 pH units.

5.1.2. TDS Meter (TDS Meter V1.0 Module)

The TDS measurement system connected the conductivity probe to the TDS Meter V1.0 module, which outputted a 0-3.3V analog signal directly to ESP32 GPIO35. GPIO35 is an input only pin from the ADC1 bank, ensuring stable analog readings since it doesn't

conflict with and isn't interfered by using the ESP's WiFi features. The TDS module internally generates a 1 kHz square wave signal to prevent electrode polarization, then converts the resulting AC signal to a DC voltage proportional to the solution's conductivity. Sampling was done by averaging multiple readings to mitigate electrical noise present in conductivity measurements.

In the integrated monitoring process, the TDS sensor measured Total Dissolved Solids concentration in parts per million, converting electrical conductivity measurement to TDS.

5.1.3. DHT22 Temperature and Humidity Sensor

The DHT22 sensor is interfaced with the ESP32 through single wire protocol connected to GPIO32, with an external 4.7 k Ω pullup resistor to 3.3V. GPIO32 is a fully capable general purpose digital I/O pin from the ADC1 group and is also free from bootstrapping constraints or RF subsystem conflicts. The sensor operates from a 3.3V supply with recommended decoupling capacitors placed near the sensor to ensure stable operation. The DHT22 comes with preloaded calibration coefficients, eliminating the need for user calibration.

The DHT22 served by providing accurate surrounding temperature and relative humidity measurements with accuracies of $\pm 0.5^{\circ}\text{C}$ across a -40°C to 80°C range and $\pm 2 - 5\%$ across 0 - 100% relative humidity.

5.1.4. ESP32-WROOM32 Development Board

ESP32-WROOM32 was chosen due to its builtin WiFi and Bluetooth capabilities, along with its high precision 12 bit resolution ADCs that supports a 0 - 3.3V input range with 11dB attenuation. The availability of these ADCs in 15+ channels makes this development board appropriate for the proposed design requiring multiple analog inputs to be taken.

5.2. Software Implementation

The software followed a client server model which made the system flexible. It consisted of firmware running on the ESP32, a backend server, a cloud database, and a frontend client application.

5.2.1. ESP32 Firmware

The ESP32 firmware was responsible for sensor data acquisition, preprocessing, and Wi-Fi communication for the transmission of data. The firmware was implemented with failsafes hence, firmware remained stable even in the face of constant monitoring.

The sensor data was periodically retrieved and sent to the cloud database via the HTTP-based communication. MicroPython was used as the firmware language, with Thonny IDE for programming and debugging. Testing and debugging were easier due to the availability of Python's REPL within the Thonny IDE.

5.2.2. Backend Server Architecture

The backend was created under the Django web framework and Django Rest Framework (DRF). The backend served as an intermediary between the database and the frontend application. It offered RESTful API endpoints to data ingestion, data retrieval, authentication and configuration.

To guarantee a secure access, JWT-based authentication was carried out. There was management of Cross-Origin Resource Sharing (CORS) that facilitated a smooth flow between the frontend and the backend.

5.2.3. Database Architecture

Supabase was employed to store sensor data with the help of a cloud-based database. It offered a scalable and dependable data storage and retrieval platform which was real time. Sensors (pH, TDS, temperature, and humidity) data was stored with the timestamps to analyze the past and evaluate the performance.

The cloud database made this data persistent, accessible across the network, and compatible with the optimization methods of the future that were based on data.

5.2.4. Frontend Architecture

The frontend application was developed using React to provide a responsive and intuitive user interface. The frontend communicated with the backend server through RESTful APIs using Axios. It displays realtime sensor readings, historical trends and current status through a dashboard.

5.3. Nutrient Solution

The nutrient solution was prepared from gathered materials which were NPK 19-19-19, Calcium Nitrate, Epsom Salt and distilled water. Two solutions A and B were prepared and stored in 500ml wash bottles. Warm distilled water was also used for quick dissolution of the materials during preparation.

Firstly, for Solution A, 120 grams of NPK 19-19-19 was dissolved into 250ml of warm distilled water. In a separate container, 60 grams of Epsom Salt was dissolved into 250ml of warm distilled water. Then both the solutions were stirred until the particles dissolved completely. After that, both 250ml solutions were mixed together and the

resultant was Solution A.

For Solution B, 180 grams of Calcium Nitrate was dissolved into 500ml of distilled water. The solution was stirred until Calcium Nitrate particles fully dissolved. After the preparation of both solutions A and B, they were transferred into 500ml wash bottles for use.

6. RESULTS AND ANALYSIS

The Automated Nutrient Dosing and Monitoring System, at the its level of its development at present, collects the data of pH, TDS, temperature, and humidity sensors and presents it on a web dashboard. The sensors are all connected to ESP32 microcontroller, which monitors realtime environmental parameters and nutrient values of the hydroponic system. ESP32 sends the data collected by the sensors to a cloud database through Wi-Fi connection, where the readings are stored in a cloud database with timestamps.

The stored data is accessed and displayed on the web dashboard so that users can monitor pH, nutrient concentration, temperature and humidity. The dashboard presents both live values and line graphs, which confirms that sensor data acquisition, wireless communication, database storage, and frontend interface are working correctly. These results demonstrate that the system is able to collect, transmit, store, and display parameters in real time. Users can also update threshold values to suit different crops.

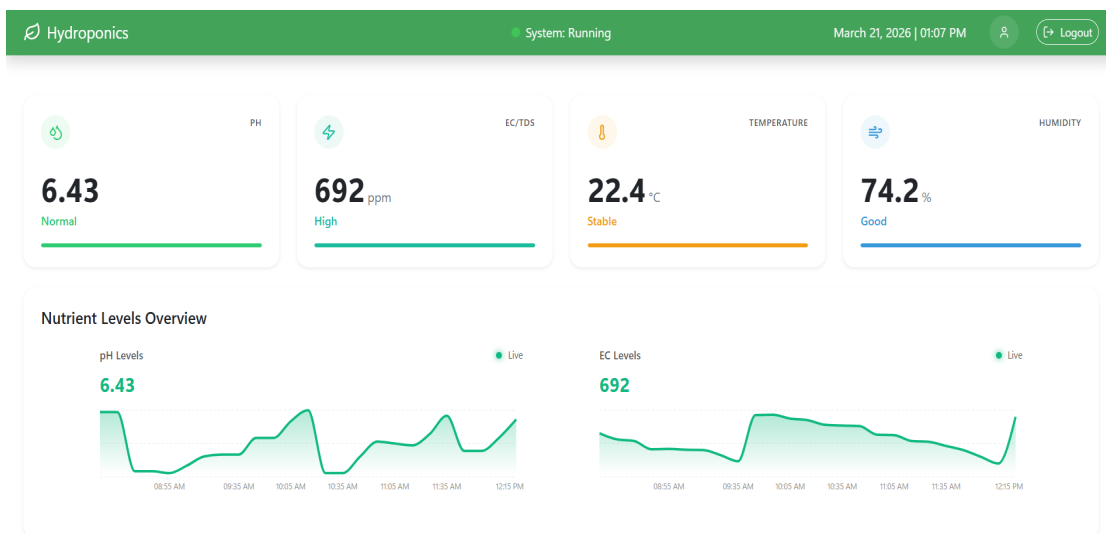


Figure 6-1. Screenshot of dashboard with current data graphs

The plant configuration page is the control interface through which users set and update the crop specific setpoints that control the automated dosing behaviour of the system. The interface is accessible through the web dashboard and communicates with the backend server through RESTful API calls. Then this updates the values to the ESP32 microcontroller through the cloud database. At the top of the page, the active plant is displayed as Lettuce, which confirms that the system is currently working under parameter ranges for a lettuce. The plant selection dropdown allows the user to switch between crop profiles, which carries its own defined thresholds for pH, EC,

and humidity. It enables the system to be adapted for different crops without requiring firmware modifications.

Plant Configuration

Select a plant and configure ideal growing conditions

Active Plant: Lettuce

Choose Plant

Lettuce

pH Level	6.2	6
EC / TDS (ppm)	650	650
Humidity (%)	70	80
pH Pump Time (ms)	270	265
TDS Pump Time (ms)	250	245
RH Fan Time (ms)	500	500

Save Configuration

Figure 6-2. Plant parameters configuration

When the operator clicks Save Configuration, the values are sent to the Supabase cloud database. From the database, ESP32 firmware fetches them at the end of each monitoring cycle as defined in the control loop. This architecture makes sure that parameter updates take effect at the next scheduled dosing check without interrupting the ongoing process. This maintains system continuity during configuration changes and provides remote access as well.

Error was observed on ESP pins GPIO4 (ADC1_CH3) and GPIO2 (ADC2_CH3) while reading analog data during testing where garbage values were observed possibly due to stray capacitances in the prototype board. GPIO2 and GPIO4 are also essential for boot

functions i.e. LOW/HIGH at boot, and bootstrapping respectively. Whenever WiFi or Bluetooth is enabled, the aforementioned pins generally referred to as ADC2 pins are taken over by the RF subsystem and are regarded as unstable for analog readings, thus they return garbage values or fail. Therefore pins ESP32 GPIO34 (ADC1_CH6), GPIO35 (ADC1_CH7), and GPIO32 pins were used for pH, TDS, and DHT22 analog data inputs respectively.

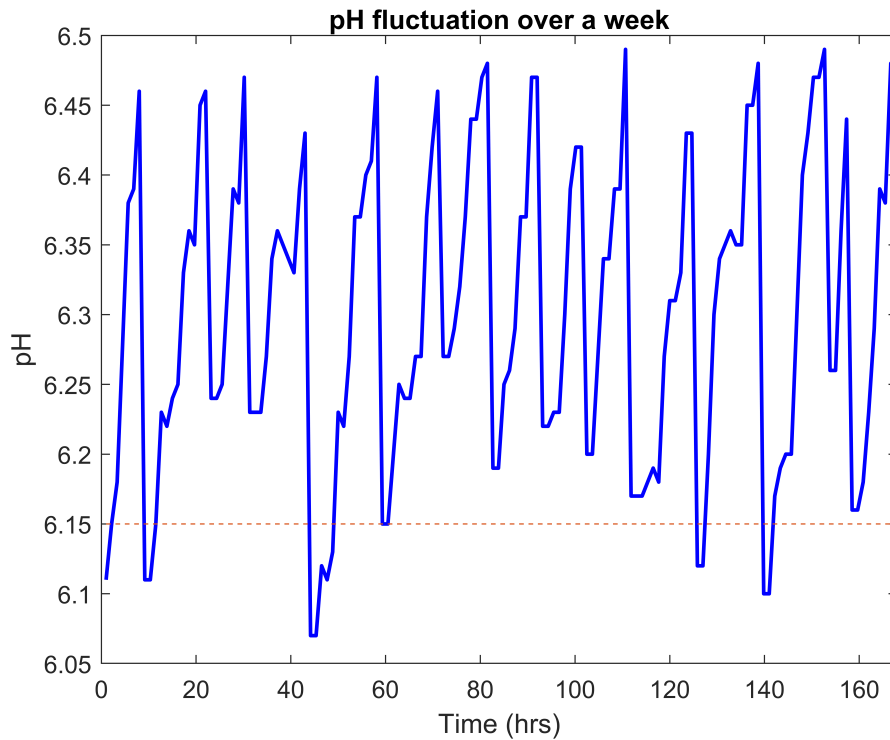


Figure 6-3. pH Fluctuation Graph over a week

The pH fluctuation graph recorded over 1 week (168-hour period) demonstrates the closed-loop control system working consistently within the optimal range of 6.0 to 6.5 for lettuce cultivation. The actual range for lettuce is 5.5 to 6.5 pH but the lower threshold of 6.0 pH was kept to avoid overdosing. Through the data, we can observe a recurring sawtooth pattern throughout the entire week, which is characteristic of threshold-based control. In each cycle, pH rises gradually due to the absorption of anions such as nitrate and phosphate by the lettuce roots as a result of nutrient dosing, which progressively alkalises the solution. When the pH reaches the upper threshold which is 6.5, the acid dosing pump is activated, which delivers a measured quantity of pH-down solution that brings the value back down to approximately 6.1 to 6.2, then after sometime the natural gradual upward drift can be seen again after the observed sharp drop which occurred due to adding acidic pH balancing solution. The amplitude of each cycle remains consistent at approximately $\pm 0.30 - 0.38$ pH units, and the correction response is applied correctly across the week without any of the values crossing the

defined boundaries.

The system did not require pH up dosing at any point during the observation period, which aligns with expected behaviour since lettuce systems under normal operating conditions tend to alkalise rather than acidify the solution. These results confirm that the pH regulation subsystem is working correctly and maintaining solution chemistry within the crop specific tolerance band.

Since the pH meter used was not a continuous measurement industrial grade equipment, the pen-type commercial pH meter had to be recalibrated rather frequently due to deviation from true pH value reflected by a spare pen-type pH meter. To mitigate inaccuracies and accidental dosing due to inaccurate pH readings, the pH meter continually submerged in the solution was recalibrated with given buffer solutions every 4 days. Along with recalibration, new values of the interpolated MID_PH_7 had to be calculated since the measured electrode potentials were also observed to deviate.

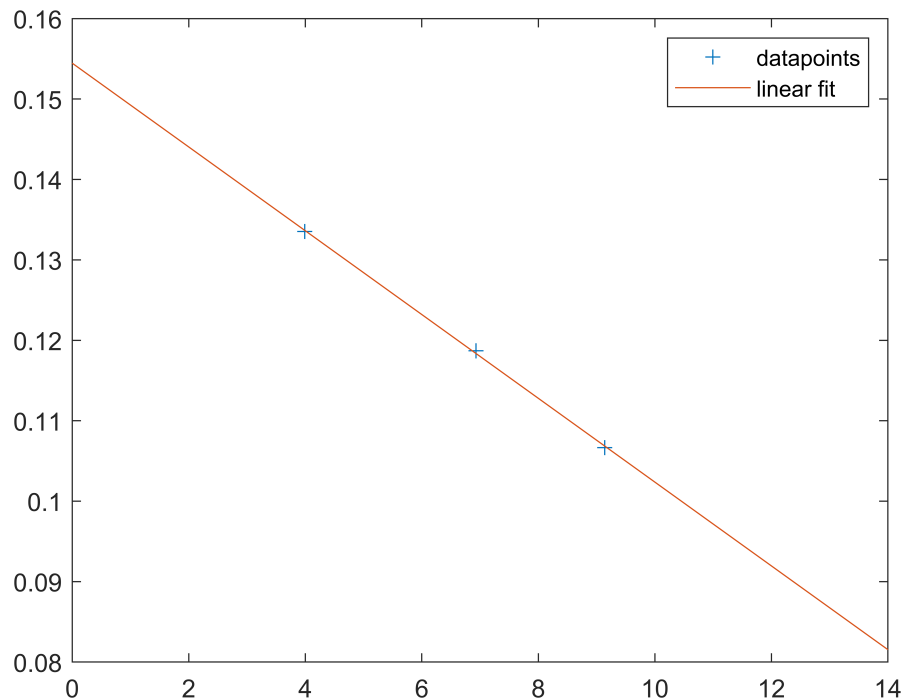


Figure 6-4. Electrode voltage vs PH interpolation

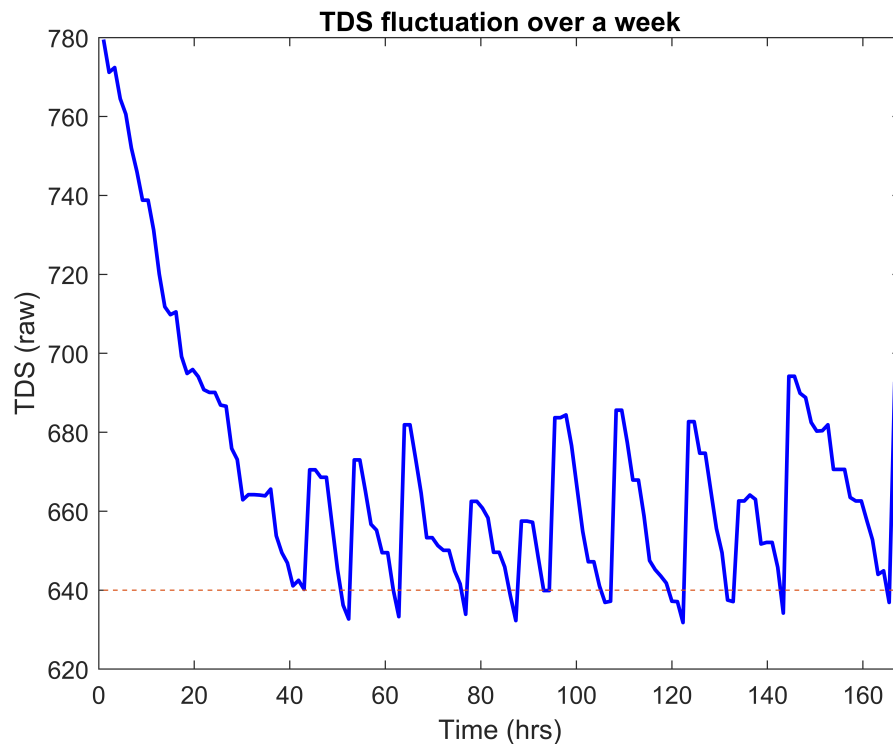


Figure 6-5. TDS Fluctuation Graph over a week

The graph above displays the TDS fluctuation over a 1 week time period showing two distinct phases of behavior. During the first 40 hour mark a steep continuous decline is observed from approximately 780 units down to 640 units, representing a drop of roughly 140 units before the first dosing correction event occurs. The decline is slightly steeper than the expected 60 to 70 units drop per day for mature stages of lettuce, and is attributed to the initial TDS value being in the higher quarter of the threshold range. From 40 hours onward, a sawtooth oscillation can be observed between the ranges of approximately 630 to 690 units, demonstrating the activation of the nutrient dosing pump to compensate for the decline caused by plant uptake and evaporation. When a dosing event occurs, it produces a sharp upward correction of 30 to 60 units. After some time, a gradual decline can be observed as nutrients are consumed by the plants. The system successfully prevents TDS from falling below the minimum threshold of 560 units throughout the observation period.

Table 6-1. System performance vs benchmark targets

S.N.	Performance Metric	Manual (Reference)	System Designed System
1	pH deviation ($\pm$ units)	0.6 - 1.2	0.30 - 0.38
2	EC deviation ($\pm$ ppm)	150 - 300	30 - 60
3	Monitoring frequency	2 - 4 $\times$ /day	144 $\times$ /day
4	pH correction time (min)	30 - 120	5 - 10
5	EC correction time (min)	30 - 120	5 - 10
6	Human labour (h/week)	4 - 8	$\sim$ 0.3
7	Sensor accuracy - pH ($\pm$)	$\pm$ 0.2 typical	$\pm$ 0.1
8	Sensor accuracy - TDS (%)	$\pm$ 5% typical	$\pm$ 2.5%

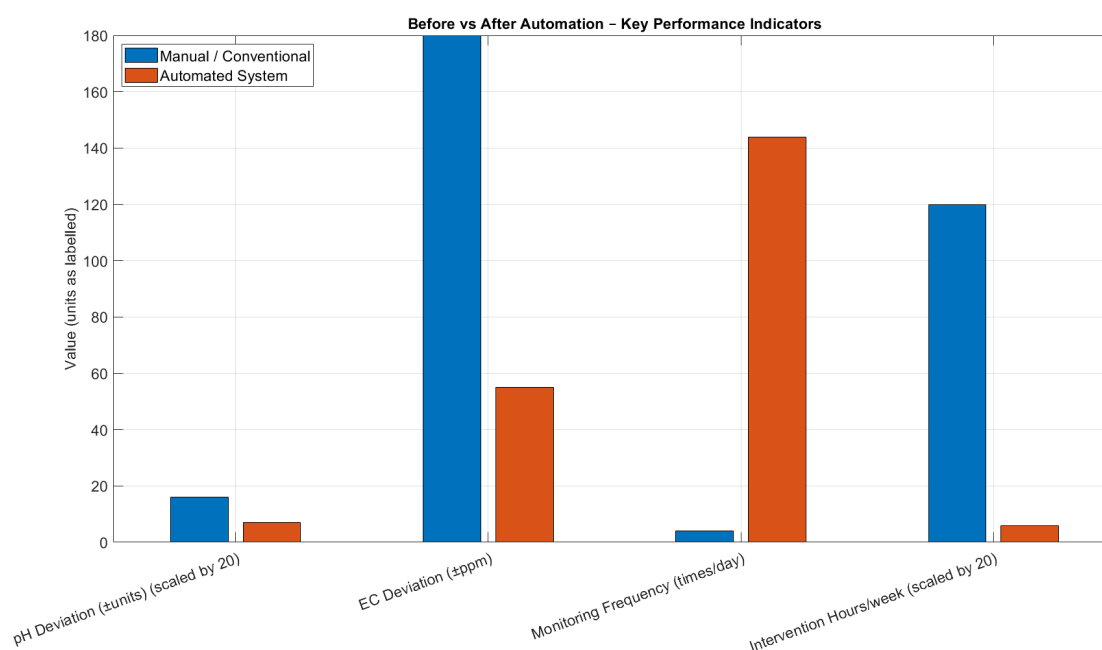


Figure 6-6. Before vs After Automation - Performance Comparison

The above bar graph shows the comparison between the performance of a hydroponics system before automation and after automation. The comparison is done across four indicators, which are pH deviation, EC deviation, monitoring frequency, and manual intervention. The automated system reduced the pH deviation from $\pm$ 0.8 units to $\pm$ 0.35 units, EC deviation from $\pm$ 180 ppm to $\pm$ 55 ppm, monitoring frequency increased from 4 to 144 checks per day as there was a 10 minute interval for checking parameters. Similarly, manual intervention was reduced from 6 hours per week to 0.3 hours per week, which was time required for calibration of sensors. Across all four metrics,

the automated approach demonstrates a clear improvement over manual operation, validating the effectiveness of the closed-loop control architecture for small scale hydroponic cultivation.

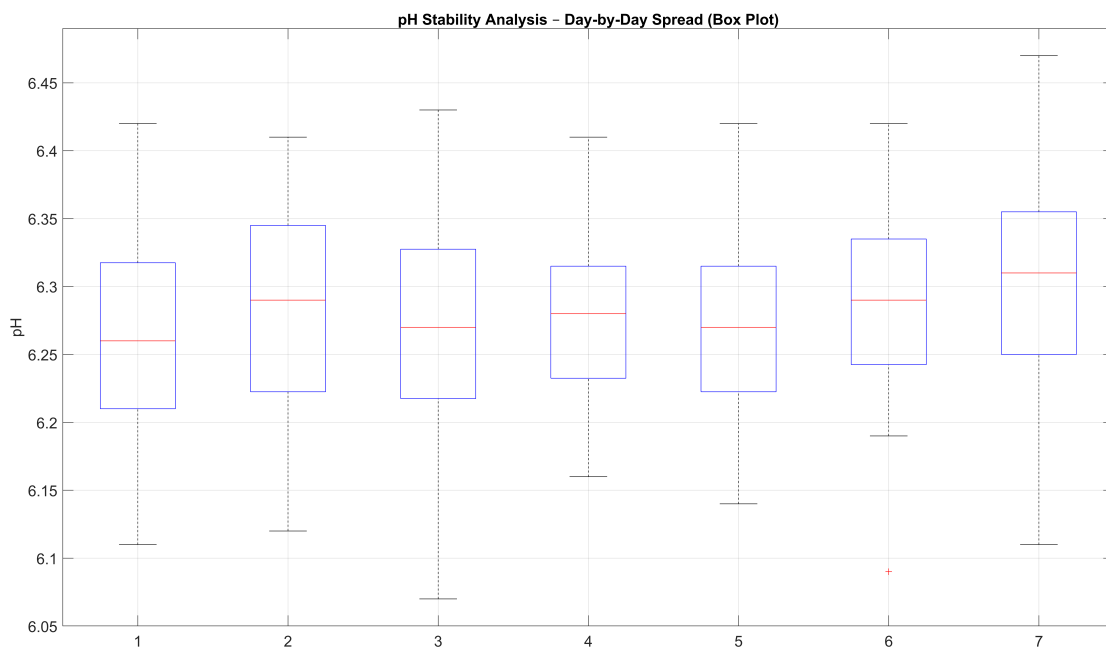


Figure 6-7. pH stability daily box plot distribution over a week

7. CONCLUSION

In conclusion, the design and development of this automated nutrient dosing and monitoring system for hydroponic crop production demonstrated the integration of embedded systems, sensing technologies, and IoT based data management in order to mitigate the drawbacks of traditional hydroponics systems. The developed system has managed to achieve a closed loop control system to control key nutrient parameters, namely pH and electrical conductivity (EC) with significantly reduced human interaction.

The utilization of an ESP32-WROOM32 microcontroller as the central processing unit allows efficient handling of realtime sensors data acquisition, control logic execution, and wireless communication. Analog sensors to measure pH and TDS and a digital DHT22 sensor for environmental monitoring ensured complete monitoring of nutrient solution chemistry and the surrounding environment.

An important advancement made by this system is the implementation of a threshold-based closed loop control strategy with stabilization delay, which is most appropriate to use in the hydroponic systems with a slow response. Compared to the traditional PID-based techniques, the offered method is simpler to compute and avoids issues such as overshoot and oscillations, making it highly suitable for low cost microcontroller platforms. The decision to regulate the EC first as opposed to pH adjustment is an indication of the view of the relationship between the concentration of the nutrient and the acidity of the solutions hence enhancing stability and effectiveness of the overall system.

Integration of IoT through Wi-Fi communication and cloud based database storage enabled real time data logging, remote monitor, and transparency of the system. The backend framework is based on Django RESTapi for safe data exchange and scalability of the system, whereas the frontend interface provides the intuitive display of sensor measurements and system conditions. This end to end architecture confirms the possibility of implementing smart hydroponic systems in real life environments with little human intervention.

Experimental observation confirms that this system is capable of continuously sensing, transmitting, storing, and displaying critical parameters, thereby establishing a stable monitoring framework. Over a 1 week(168 hour) validation period of 1 week, the system maintained pH within a deviation of ± 0.38 units and EC within ± 60 ppm of their respective setpoints, satisfying the defined requirements of ± 1.0 pH units and ± 200 ppm respectively. Monitoring frequency increased from approximately 4 manual checks per day to 144 automated readings per day, while operator intervention

was reduced from an estimated 4 - 8 hours per week to approximately 0.3 hours per week. Corrective response for both pH and EC deviations was consistently achieved within 5 - 10 minutes of threshold breach, and the total hardware cost of NPR 15,170 remained within the budget constraint. Some hardware limitations like the instability of ADC pins and sensor drift during testing were identified. The most significant of these was the progressive drift of the commercial pen-type pH electrode under continuous immersion, which necessitated manual recalibration every four days and poses the primary reliability risk for long term unattended operation of this whole system. Additionally, the fixed 10 minute polling interval introduces a maximum detection latency of 10 minutes per deviation event, which may be insufficient for faster responding crops or smaller reservoir volumes. These issues can be addressed in the optimization of the system in the future through better hardware design and improved calibration processes.

Despite the limitation of being a single crop operation and manually calibrated pH sensor, the system has great potential of being extended further in the future. Multi crop profiles, adaptive control strategies, and machine learning or even RL based predictive models would transform the system to a proactive control framework. Also, incorporating greater sensor redundancy and fault detection, alongside a mobile user interface, would further improve system reliability and usability.

Overall, this project demonstrates a scalable, cost-effective, and technically feasible approach to precision hydroponic agriculture, with a validated hardware cost of NPR 15,170 well within the targeted budget.

8. FUTURE ENHANCEMENTS

8.1. Multi-Crop Support and Adaptive Setpoints

The designed hydroponics system is limited to a single crop which is lettuce (*Lactuca sativa*). At the present state of the system, the crop-specific setpoints such as the pH and EC ranges as well as the parameters including pump time period and fan runtime are only available for lettuce. A crop database with predefined parameter ranges, along with automatic profile switching based on the selected crop, can be implemented. In hydroponic cultivation, nutrient requirements can vary across different growth phases such as seedling, flowering, fruiting, vegetative, and harvest stages. So, an adaptive nutrient management system can be added based on the growth stage of the crop.

8.2. Integration of Machine Learning for Predictive Control

The current system is a closed-loop system, resulting in the system being reactive to the deviation that has occurred. Machine learning enables predictive control, where the deviation is anticipated prior to the occurrence. The key implementations could be nutrient consumption prediction, intelligent dosing, anomaly detection, and more. Using historical EC data, the system can predict how long it takes for nutrient levels to drop below the threshold, enabling dynamic dosing intervals. Instead of fixed dosing durations, the relationship between EC/pH change and pump activation time can be modelled to enable adaptive, non-linear dosing. When an anomaly such as gradual sensor drift occurs it can be detected easily or when sudden abnormal readings occur, they can be flagged as a fault in the system.

8.3. Improved Sensor Accuracy and Calibration Mechanisms

The current system can only be manually calibrated since there is sensor drift with time. The improvements to be made in the future can be on automatic calibration, more precise industrial grade sensors, and redundancy of sensors to provide fault tolerance. This framework will self-check by using solutions known to have pH levels, such as pH 4, 7, and 10. It will test the pH of such solutions, and compare the measurements with the expected one and adjust itself, as needed. More than one sensor can also be used to measure the same parameter, such as two pH probes. In this manner, comparisons can be made between the outcomes of the two sensors and identify when one of them is not functioning properly. In case results are not consistent, the system will ignore such data. Lastly, by substituting cheap hobby sensors with high quality probes, it is possible to improve precision, stability, and dependability and make the system applicable to a long term and real life use.

8.4. Mobile Application and User Interface Enhancement

The current system has a web dashboard which displays data and allows users to make changes to the threshold and dosing options. To increase user interaction, a mobile application is an alternative to the dashboard. The mobile application will provide monitoring of key parameters such as pH, EC, temperature, and humidity in realtime through an user friendly interface. There will be an additional feature which sends alerts to keep users informed in realtime when parameters deviate from the predefined thresholds which was not available in the web dashboard. The application can also aid the operation of remote controls, where setpoints can be adjusted, crop profiles updated, or pumps and fans, operated manually from any place. Hence, the development of a mobile application improves usability and user-friendly as well as more applicable to real life and large scale applications.

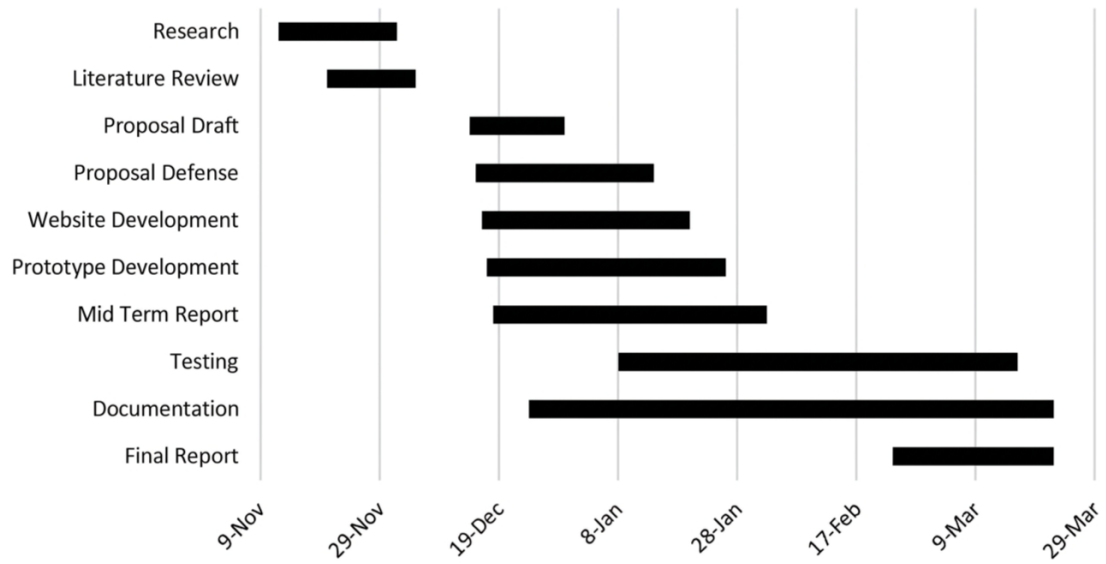
A. PROJECT BUDGET

Table A-1. Project Budget

S.N.	Item Name	Unit Price (NPR)	Quantity	Total (NPR)
1	TDS sensor	1,500	1	1,500
2	pH Meter	700	1	700
3	DHT22 sensor	300	1	300
4	ESP32-WROOM32	900	1	900
5	Submersible water pump	1,500	1	1,500
6	UPVC pipe 75mm (Pipe and elbow sockets)	2,000	1	2,000
7	CPVC pipe 0.75inch (Pipe and elbow sockets)	3,000	1	3,000
8	IRL2203N MOSFET	50	5	250
9	Matrix Board	250	1	250
10	1N5822 Schottky Diode	35	5	175
11	5V DC Water Pump	250	3	750
12	DC Adapter AC To DC 12V 1.5A	350	1	350
13	LM2596 DC-DC buck converter	250	1	250
14	12V DC Fan	200	1	200
15	Calcium Nitrate	500	1	500
16	Epsom Salt (200gm)	150	1	150
17	NPK (19:19:19) (100gm)	100	3	300
18	Wash Bottle	125	3	375
19	Distilled Water (in liters)	25	4	100
20	Copper Clad Board	90	2	180
21	Plastic Cover (in m ²)	60	4	240
22	Heat Shrink Tube (in m)	200	1	200
23	Miscellaneous	-	-	1000
Grand Total				15,170.0

B. PROJECT SCHEDULE

Table B-1. Gantt Chart of the project



C. SCHEMATIC DIAGRAM OF TDS SENSOR

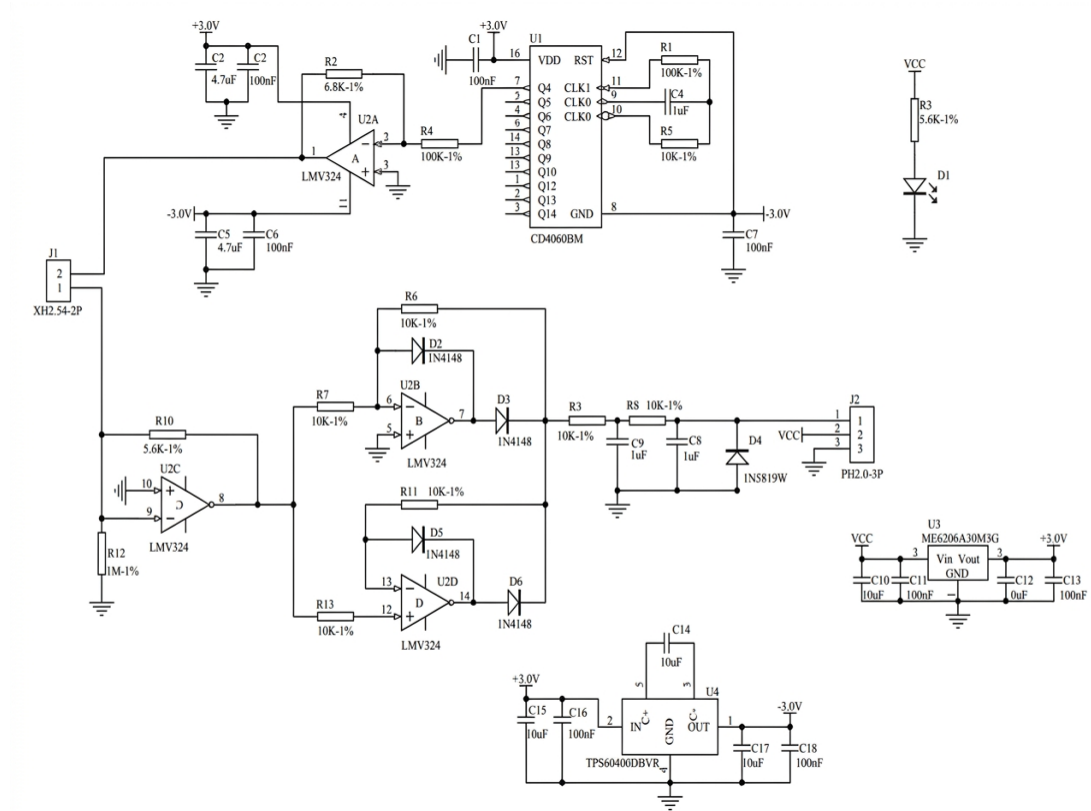


Figure C-1. TDS Sensor Schematic Diagram

D. SCHEMATIC DIAGRAM OF ESP32

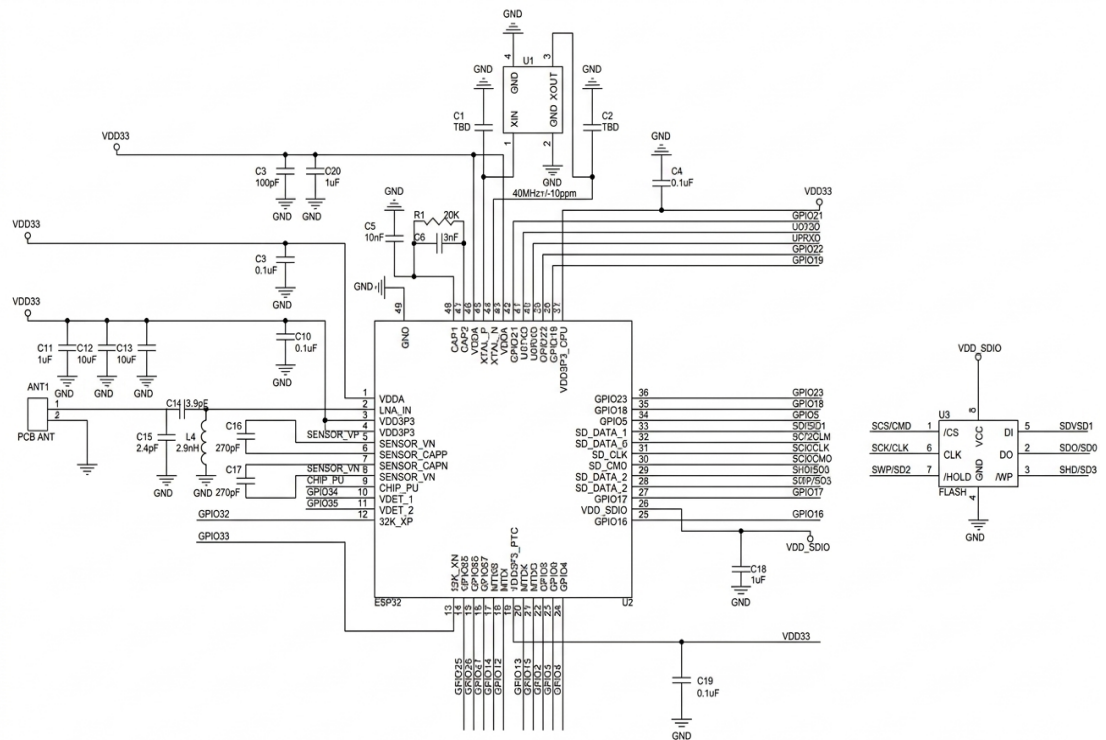


Figure D-1. ESP32 Schematic diagram

E. CIRCUIT DIAGRAM

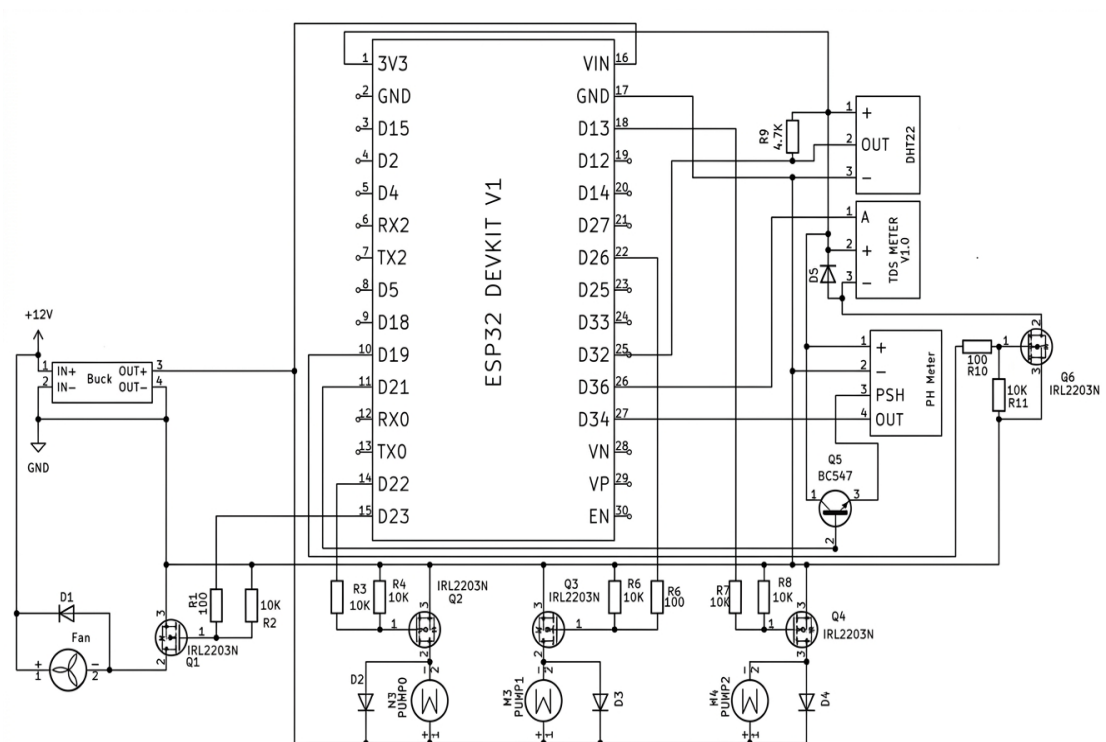


Figure E-1. Circuit Diagram of the system

F. PCB DESIGN

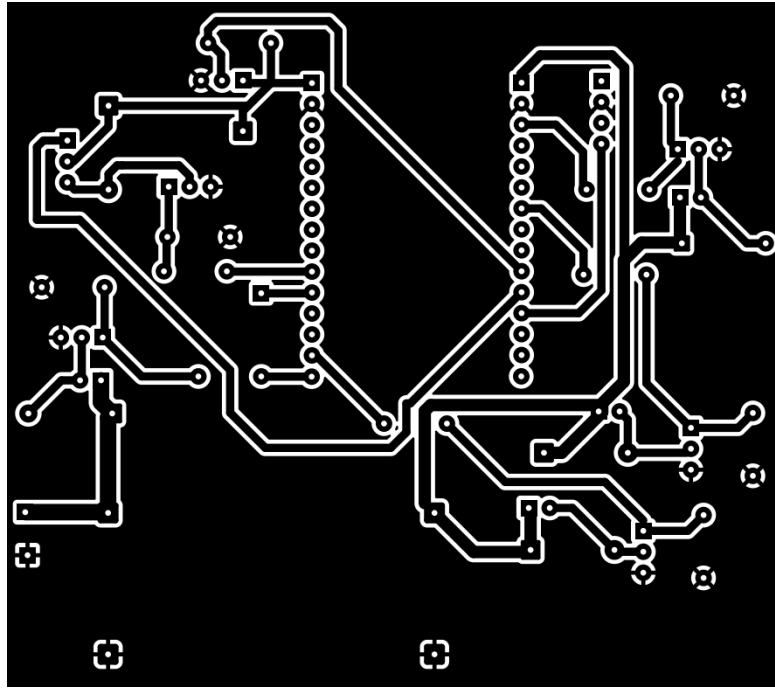


Figure F-1. PCB Design of the system

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